

Predicting Probability of Default using Logistic Regression Approach

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Abstract

This study uses logistic regression model of Ohlson (1980) for the prediction of probability of default in Indian context using accounting and market data from 2008-09 to 2012-13 of small sample of 40 firms to figure out the effectiveness of logistic model over different time horizons. The study finds that the regression model works better for shorter duration than that of the longer duration. Also even if most of the variables are of the models are not significant, the classification capability of models are very high in range of 95% to 96.7%

Keywords: Default Probability, LOGIT, Logistic Regression, Financial Distress, Credit Scoring

Introduction

Since times when the practice of lending started, lenders have been worried about the borrowers' ability to return the money back. Different ways and methods have been evolved and used to figure it out and in the process the modern practices of credit rating and default, internal as well as external, has evolved. The importance of credit rating and default has increased significantly with the development of organized banking and financial industry as the whole economy now considerably depends on it. So the prediction of default has become very crucial to banks, financial institutions as well as regulators as the cost of default is very high for the whole economic system.

Modern banking system in India started to develop in early 18th Century in 1720 with Bank of Bombay, so the market for debt (RBI, 2017). Like other economies Indian banking system also had been worried about default prediction and has been depending on external and internal credit rating systems. But in academics very little work has been carried out in Indian context.

This study uses logistic regression method to arrive upon a credit scoring model as well as probability of default in Indian context using accounting and market data from 2007-08 to 2012-13 of small sample of 40 firms to figure out the effectiveness of logistic model over different time horizons and is divided into three parts. Model 1 is developed for two years data, Model 2 is developed for three years data and the Model 3 is developed for 6 years data.

Review of Literature

Since the first systematic study of credit scoring models and default prediction of FitzPatrick (1932), different methods and models have evolved and thousands of studies across the world

have been performed using different data sets and variables depending upon respective economic realities.

Beaver (1966) exhibited the importance of different financial ratios in prediction of failure of the firms as well as role credit extension to firms used by banking and financial sector (Beaver, 1966). Theory says that when the value of assets is lower than that of the liabilities or because of lack of liquidity, firms tend to default (Wilcox, 1971). A few financial and structural ratios can help to understand the same (Scott, 1981). Altman (1968) proposed a multiple discriminant analysis based Z-score model that used different financial and market ratios to arrive at Z-score which could provide information about the possibility of default by the firm in near future. The model used ratios and the accuracy of model was 94%. This changed whole arena of default prediction completely. The results could be used for loan evaluation by banking sectors as well as internal control and investment criteria. (Altman E. I., 1968).

Bandyopadhyay (2006) developed a Z-score model which exhibited high predictive power in detecting possible defaulting firms. The developed model outperformed the original Altman's model in Indian context. The new Z-score model can provide early signals about the firms financial positions. The analysis of logit model clearly indicates that inclusion of financial and non-financial parameters improves the predictive ability of the model. The default probability calculated using Logit can be helpful in deciding credit risk capital as well pricing of debt (Bandyopadhyay, 2006).

Liang (2003) conducted a comparative study on Chinese firm between MDA and logistic regression model exhibiting that logistic regression providing relatively higher predictive ability than that of the MDA (Liang, 2003). A comparison of ZETA model and Z-score model ZETA model provide improved accuracy. Reasons may be the more relevant data and

larger numbers of industrial firms used in the analysis for ZETA analysis (Altman E. I., 2000). The information from structural models can improve the accuracy in prediction in credit scoring models like logistic regression model. Financial statements information along with firm specific information improve the predictive power of the model (Bandyopadhyay, 2007). Market value of the assets and volatility in the value of assets are important factors in the process of arriving at default probability and have inverse relationship with default probability (Upadhyay, Sharma, & Singh, 2014).

Objectives

From Basel to RBI guidelines to internal assessment to external credit rating, the number of reported corporate debt default is on rise. Banks and Corporate Affairs Ministry are regularly issuing list of defaulters. The present study uses logistic regression method to arrive upon a credit scoring model as well as probability of default in Indian context using accounting and market data from 2007-08 to 2012-13 of small sample of 40 firms to figure out the effectiveness of logistic model over different time horizons

Data and Methods

For the purpose of study 40 firms listed on Bombay Stock Exchange have been randomly selected of which 16 firms have defaulted at least once during the six year period from 2007-08 to 2012-13 and 24 firms are non-defaulting companies from different industries. To predict probability of default as well as credit scoring models over different periods using logistic regression SPSS software has been used. The study has used the same variables used by Ohlson (1980) using the below model (Ohlson, 1980):

$$\text{O-Score} = \alpha - \beta_1 \text{WC/TA} + \beta_2 \text{TL/TA} + \beta_3 \text{CL/CA} + \beta_4 \text{NI/TA} + \beta_5 \text{FFO/TL} + \beta_6 \text{CHIN} + \beta_7 \text{Size} + \beta_8 \text{X} + \beta_9 \text{Y}$$

Where

WC/TA = Working capital to total assets ratio

TL/TA = Total liabilities to total assets ratio

CL/CA = Current liabilities to current assets ratio

NI/TA = Net income to total assets ratio

FFO/TL = Funds from operations to total liabilities

CHIN = $(NI_t - NI_{t-1}) / (NI_{t-1} + NI_{t-2})$, where NI_t is net income for the most recent period. The denominator acts as a level indicator. The variable is thus intended to measure change in net income

Size = Log (TA/GNP price-level index)

$X = 1$ if $TL > TA$, 0 otherwise

$Y = 1$ if a net loss for the last two years, 0 otherwise

From the analysis the following three models has been found.

Model 1 (Two Years Data):

$$O\text{-Score} = -7.841 + 10.802 \text{ WC/TA} + 11.224 \text{ TL/TA} + 3.33 \text{ CL/CA} - 23.459 \text{ NI/TA} + 7.097 \\ \text{FFO/TL} - 9.196 \text{ CHIN} - 0.82 \text{ Size} - 24.162 X + 27.014 Y$$

Model 2 (Three Years Data):

$$O\text{-Score} = -3.205 + 7.518 \text{ WC/TA} + 3.408 \text{ TL/TA} + 2.027 \text{ CL/CA} - 85.666 \text{ NI/TA} - 2.567 \\ \text{FFO/TL} - 1.928 \text{ CHIN} - 0.724 \text{ Size} + 5.991 X + 5.528 Y$$

Model 3 (Six Years Data):

$$O\text{-Score} = -9.238 - 0.062 WC/TA + 1.637 TL/TA - 0.927 CL/CA - 3.099 NI/TA - 16.866 \\ FFO/TL - 0.465 CHIN + 0.652 Size + 19.093 X + 4.696 Y$$

Data Analysis

This section is divided into three parts for the three models developed and data has been analyzed accordingly. For the classification, the cut-off value is taken as 0.5 for all the three models.

Model 1 (Two Years Data)

Model 1 has been developed using latest two years data that is from 2011-12 to 2012-13. Table 1.1 provides the model summary of the developed Model 1. Cox & Snell R Square which is considered as pseudo R square for logistic regression method is 0.638. That means 63.8% of total variation in the default prediction can be explained by this model. Nagelkerke R Square which is considered as adjusted R Square in logistic regression method is 0.891. That means this model can explain 89.1% of default incidents on the basis of used variables in the analysis.

Table 1.1: Model Summary			
Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	19.551	0.638	0.891

From table 1.2, the intercept is -0.731. The Standard Error is 0.239, not far away from 0. This can be considered to be good. Wald Chi Square test tests the null hypothesis that the constant equals to 0. The significance value is 0.002 which is lower than the critical value 0.05. Hence null hypothesis is rejected. That means constant is not zero.

Table 1.2: Variables in the Equation						
	B	S.E.	Wald	df	Sig.	Exp(B)
Constant	-0.731	0.239	9.375	1	0.002	0.481

From table 1.3, only one variable that is total liabilities to total assets ratio is found to be significant in this developed model and all the other variables are found to be insignificant.

Table 1.3: Variables in Equation					
	B	S.E.	Wald	df	Sig.
WCTA	10.802	5.793	3.477	1	.062
TLTA	11.224	5.452	4.238	1	.040
CLCA	3.330	3.058	1.186	1	.276
NITA	-23.459	50.947	.212	1	.645
FFOTL	7.097	10.918	.423	1	.516
CHIN	-9.196	5.031	3.342	1	.068
Size	-.820	.750	1.196	1	.274
X(1)	-24.162	22131.282	.000	1	.999
Y(1)	27.014	3643.421	.000	1	.994
Constant	-7.814	6.225	1.575	1	.209

From classification table 1.4, it found that this model can predict default accurately by 95%. Only 5% of the total predictions are wrongly predicted. Type 1 error is found to be 1.25% while type 2 error is found to be 3.75% making this model strong.

Table 1.4: Classification Table			
Observed	Predicted		
	Z		Percentage Correct
	No Default	Default	
No Default	53	1	98.1
Default	3	23	88.5
Overall Percentage			95.0

Model 2 (Three Years Data)

Model 2 has been developed using latest three years data that is from 2010-11 to 2012-13. Table 2.1 provides the model summary of the developed Model 2. Cox & Snell R Square which is considered as pseudo R square for logistic regression method is 0.588. That means 58.8% of total variation in the default prediction can be explained by this model. Nagelkerke R Square which is considered as adjusted R Square in logistic regression method is 0.878. That means this model can explain 87.8% of default incidents on the basis of used variables in the analysis.

Table 2.1: Model Summary		
-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
26.430	0.588	0.878

From table 2.2, the intercept is -1.144. The Standard Error is 0.213, not far away from 0. This can be considered to be good and is better than the Model 1. Wald Chi Square test tests the null hypothesis that the constant equals to 0. The significance value is 0.00 which is lower than the critical value 0.05. Hence null hypothesis is rejected. That means constant is not zero.

Table 2.2: Variables in the Equation						
	B	S.E.	Wald	df	Sig.	Exp(B)
Constant	-1.144	0.213	28.759	1	.000	.319

From table 2.3, only one variable that is X variable is found to be significant in this developed model and all the other variables are found to be insignificant.

Table 2.3: Variables in the Equation					
	B	S.E.	Wald	df	Sig.
WCTA	7.518	4.537	2.747	1	0.097
TLTA	3.408	5.298	.414	1	0.520
CLCA	2.027	2.530	.642	1	0.423
NITA	-85.666	48.190	3.160	1	0.075
FFOTL	-2.567	16.965	.023	1	0.880
CHIN	-1.928	1.851	1.086	1	0.297
Size	-.724	.715	1.024	1	0.312
X			4.071	2	0.131
X(1)	5.991	2.970	4.070	1	0.044
X(2)	5.528	3.018	3.356	1	0.067
Constant	-3.205	8.055	.158	1	0.691

From classification table 2.4, it found that this model can predict default accurately by 96.7%. Only 3.3% of the total predictions are wrongly predicted. Type 1 error is found to be less than 1% while type 2 error is found to be 2.5% making this model strong.

Table 2.4: Classification Table			
Observed Z	Predicted		
	Z		Percentage Correct
	No Default	Default	
No Default	90	1	98.9
Default	3	26	89.7
Overall Percentage			96.7

Model 3 (Six Years Data)

Model 3 has been developed using latest two years data that is from 2007-08 to 2012-13. Table 3.1 provides the model summary of the developed Model 3. Cox & Snell R Square is 0.39. That means 39% of total variation in the default prediction can be explained by this model. Nagelkerke R Square is 0.699. That means this model can explain 69.9% of default incidents on the basis of used variables in the analysis.

Table 3.1: Model Summary		
-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
77.216	0.390	0.699

From table 3.2, the intercept is -1.802. The Standard Error is 0.185, not far away from 0. This can be considered to be good and is better than Model 1 and Model 2. Wald Chi Square test tests the null hypothesis that the constant equals to 0. The significance value is 0.000 which is lower than the critical value 0.05. Hence null hypothesis is rejected. That means constant is not zero.

Table 3.2: Variables in the Equation						
	B	S.E.	Wald	df	Sig.	Exp(B)
Constant	-1.802	0.185	94.713	1	0.000	0.165

From table 3.3, only one variable that is Y and constant are found to be significant in this developed model and all the other variables are found to be insignificant.

Table 3.3: Variables in the Equation					
	B	S.E.	Wald	df	Sig.
WCTA	-.062	1.286	.002	1	.962
TLTA	1.637	.966	2.871	1	.090
CLCA	-.927	.604	2.356	1	.125
NITA	3.099	21741.028	.000	1	1.000
FFOTL	-16.866	6550.188	.000	1	.998
CHIN	-.465	.391	1.412	1	.235
Size	.652	.355	3.382	1	.066
X(1)	19.093	16589.586	.000	1	.999
Y(1)	4.696	.766	37.578	1	.000
Constant	-9.238	3.385	7.447	1	.006

From classification table 3.4, it found that this model can predict default accurately by 95%. Only 5% of the total predictions are wrongly predicted. Type 1 error is found to be 1.66% while type 2 error is found to be 3.33% making this model strong.

Table 3.4: Classification Table			
Observed Z	Predicted		
	Z		Percentage Correct
	No Default	Default	
No Default	202	4	98.1
Default	8	26	76.5
Overall Percentage			95.0

Interpretation and Implications

From the Model Summary tables for three models, it is clear that Model 1 for two years data performs better than that of the two other models. Also it can be said that smaller period is better to have a better and robust model as it can be seen that with increasing number of years, models' ability falls. From Variables in Equation tables, it can be said that there is not much difference in the three models. From Second tables for each model Variables in Equation, it is clear that all the three models are not robust. Only one variable for each model are found to be significant and constant for the Model 3 is found to be significant.

From the classification tables of the three models, Model 1 and Model 3 can equally predict correctly at 95% with almost similar level of type 1 and type 2 errors. But the Model 2 outperforms the other two models as it can accurately predict 96.7% with less than 1% of type 1 error and 2.5% of type 2 error.

Conclusion

On the basis of the above analysis it can be said that the Model 1 is the most robust model but on the basis of the classification, Model 2 is found to be better. Also form the overall analysis, it can be concluded that even if most of the variable in the three models are not

found to be significant, the models can classify accurately in a range of 95% to 96.7% which can be considered to be high and comparable to many studies like Ohlson (1980) and Bandyopadhyay (2006). So it can be concluded that logistic regression method works better over a shorter duration than the longer duration as the results of the study suggests.

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